



Agentic AI HOW?

Framework_(s)

1

OpenAI agent
SDK + Raw
Python

2

OpenAI agent
SDK in
Grasshopper
/Rhino

3

FastMCP >
MCPserver +
Any LLM UI

4

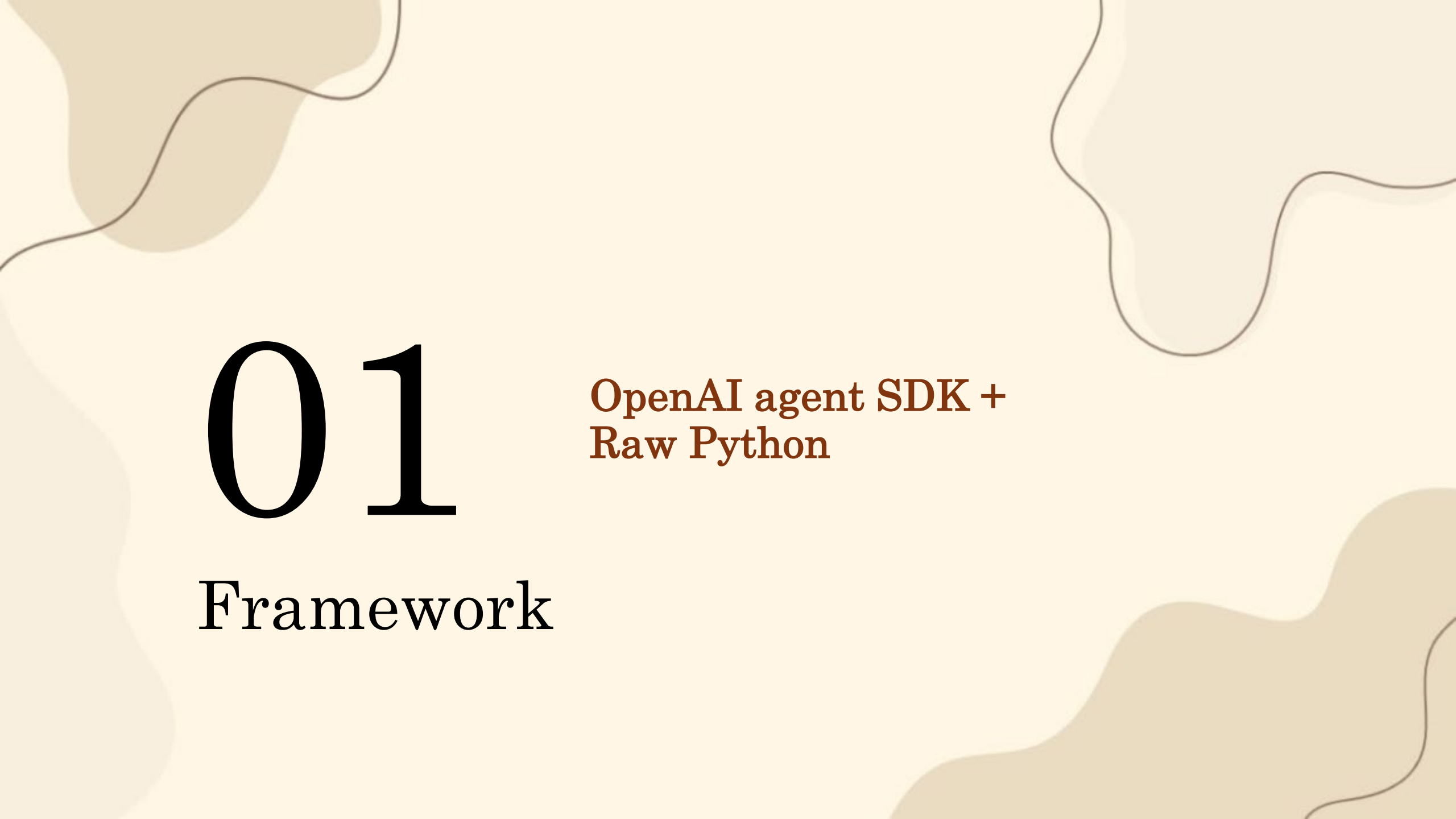
FastMCP >
MCPserver +
Any LLM UI +
RhinoCompute

5

FastMCP >
MCPserver +
Any LLM UI +
RhinoCompute
+ Strands
Agent SDK +
Streamlit
ChatApp

6

Grasshopper +
Swiftlet + Any
LLM UI



01

Framework

OpenAI agent SDK +
Raw Python

Framework 01

```
tools_example.py > ...
1  from agents import Agent, Runner, SQLiteSession, WebSearchTool, ModelSettings, function_tool, FileSearchTool
2
3
4  # Create a session to store the agent's interactions
5  session = SQLiteSession(
6      session_id="test_session",
7      db_path="agent_sessions.sqlite")
8
9
10 # Define a function tool to move a point in 3D space
11 @function_tool
12 async def move_point(x: float, y: float, z: float) -> str:
13     """
14     Move a point in 3D space and return the length of the line segment between the original point and the new point.
15
16     Args:
17         x (float): The x-coordinate of the point.
18         y (float): The y-coordinate of the point.
19         z (float): The z-coordinate of the point.
20     """
21
22     point_01 = [5,5,5]
23     point_02 = [x,y,z]
24     distance = ((point_02[0] - point_01[0]) ** 2 + (point_02[1] - point_01[1]) ** 2 + (point_02[2] - point_01[2]) ** 2) ** 0.5
25     return f"Moved point from {point_01} to {point_02}. Line length is {distance} units."
26
27 # Create an agent with the move_point tool
28 test_agent = Agent(
29     name="Test Agent",
30     model="gpt-5",
31     instructions="You are a helpful agent that moves points in 3D space to meet user requirements.",
32     tools=[move_point]
33 )
34
35 # Run the agent with a prompt to move the point in the Y direction
36 prompt = "Move the point in Y direction until the length of the line is at least 45 units."
37 result = Runner.run_sync(test_agent, prompt, session=session, max_turns=10)
38
39 print(result.final_output)
```

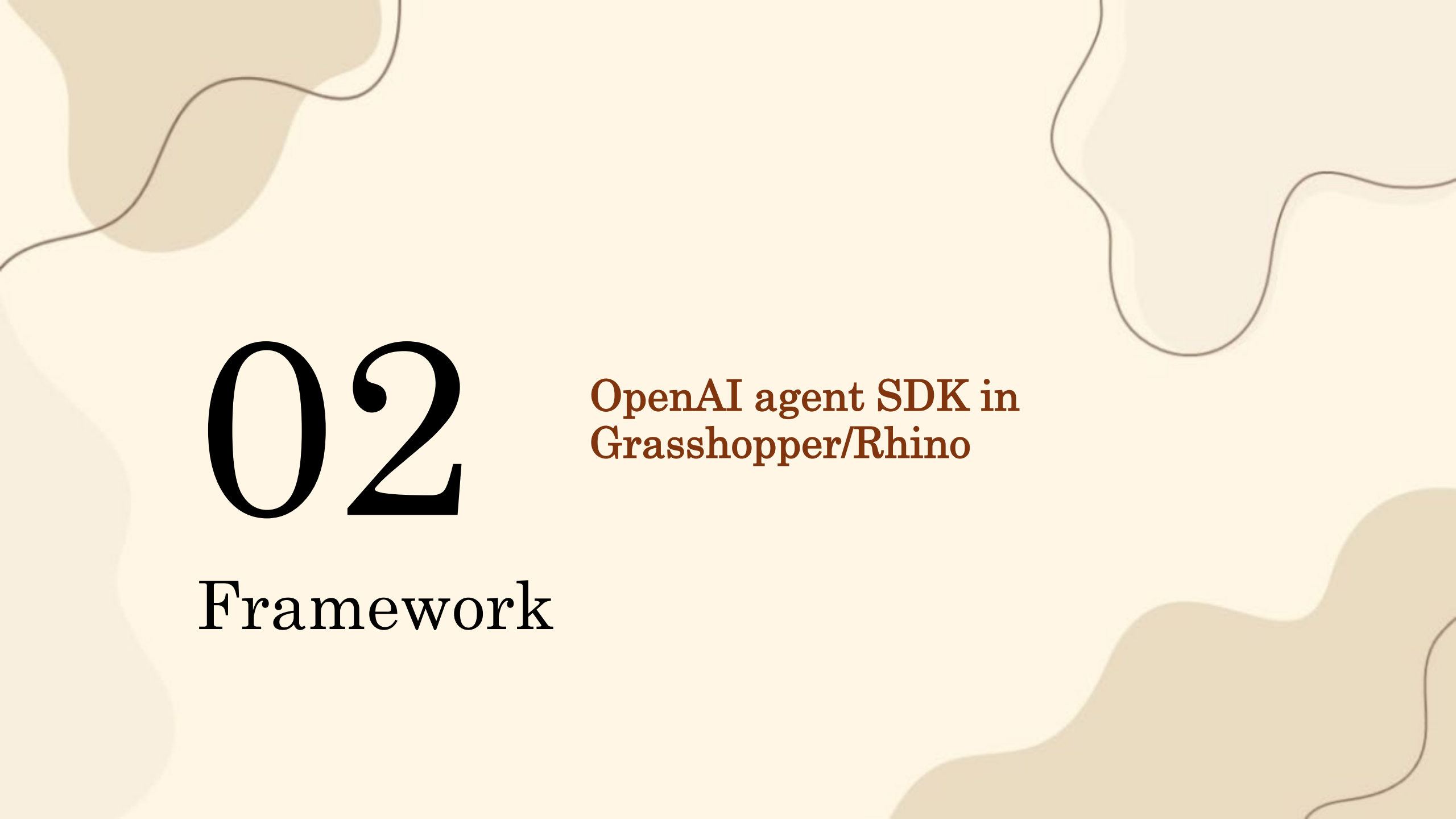
Framework *01*

Merits

- Learn about agentic workflow
- Explore tools available in OpenAI SDK
- Concentrated and fast learning

Demerits

- No CAD environment connected
- Need Basic programming background.
- Limiting broader scopes and understanding proper use cases.
- No learning about MCP (Model context protocol)

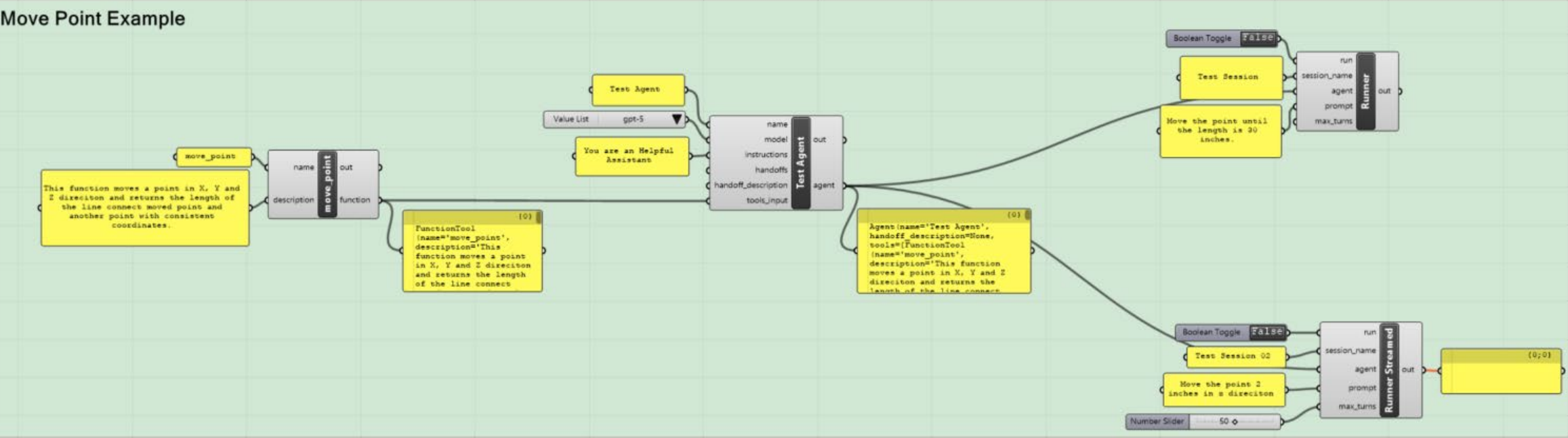


02

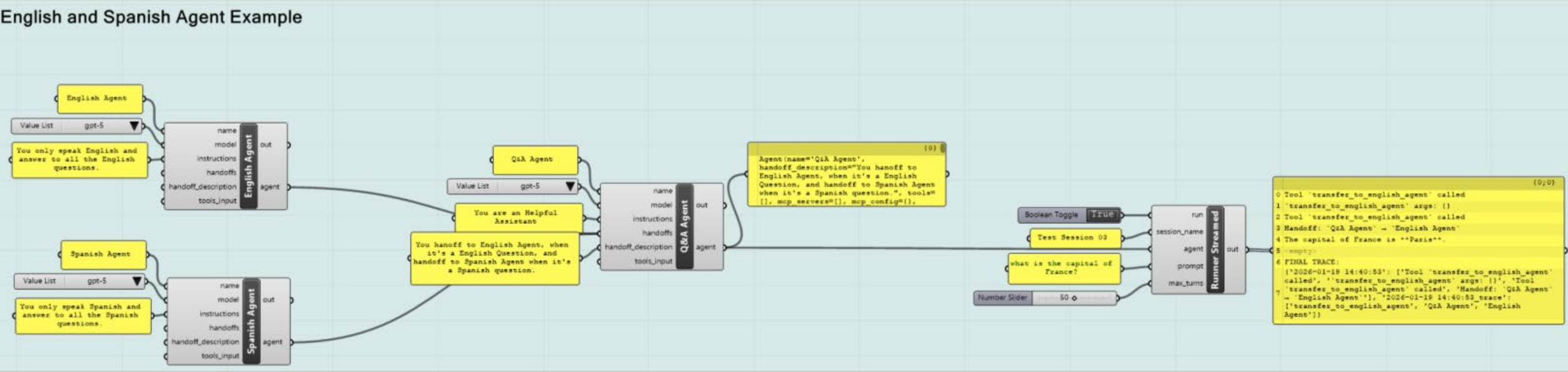
Framework

OpenAI agent SDK in
Grasshopper/Rhino

Move Point Example



English and Spanish Agent Example



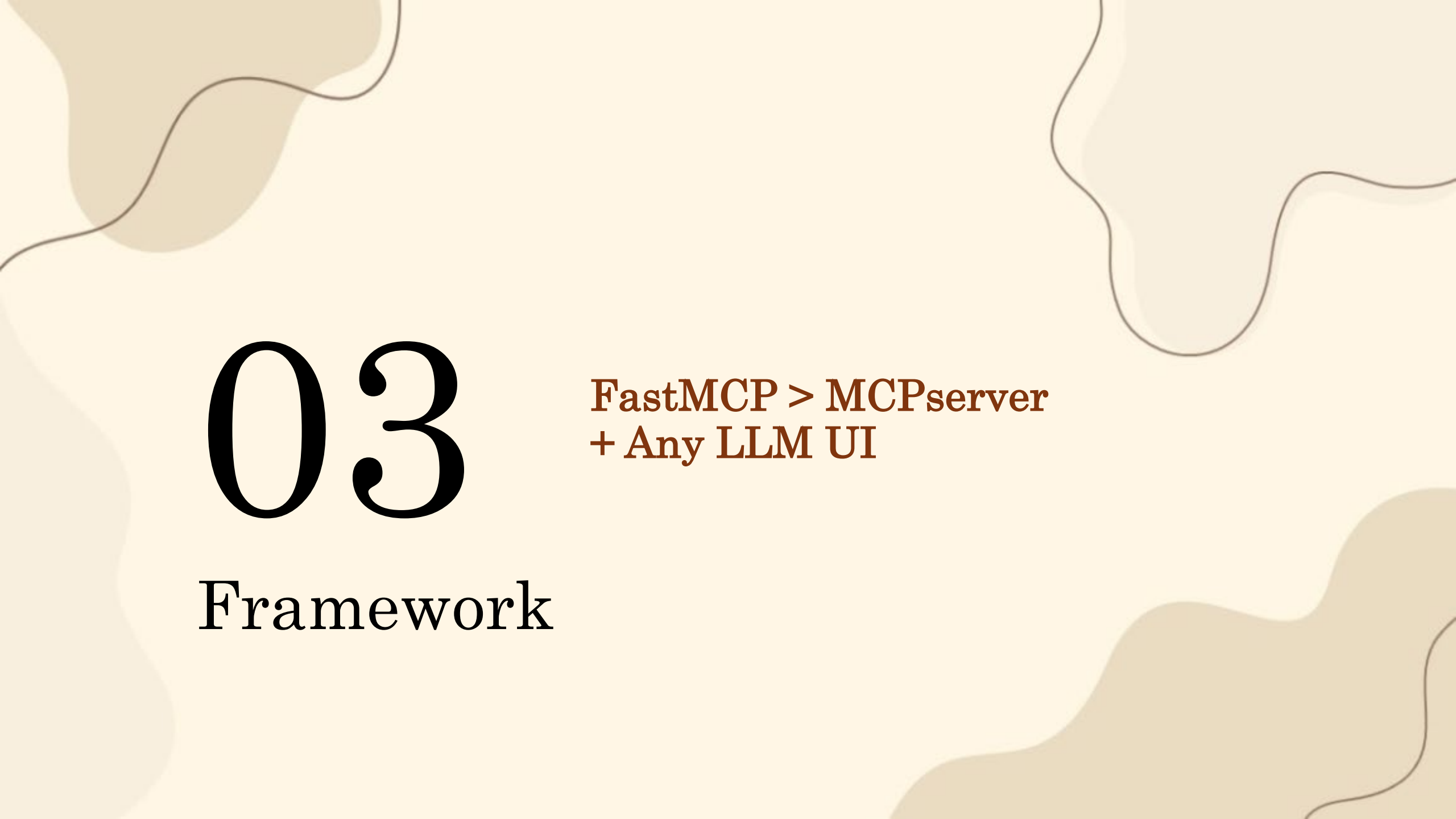
Framework *02*

Merits

- Can be taught to students have no Coding Background
 - Minimum coding requirements for creating functions only (Students can use LLMs for that).

Demerits

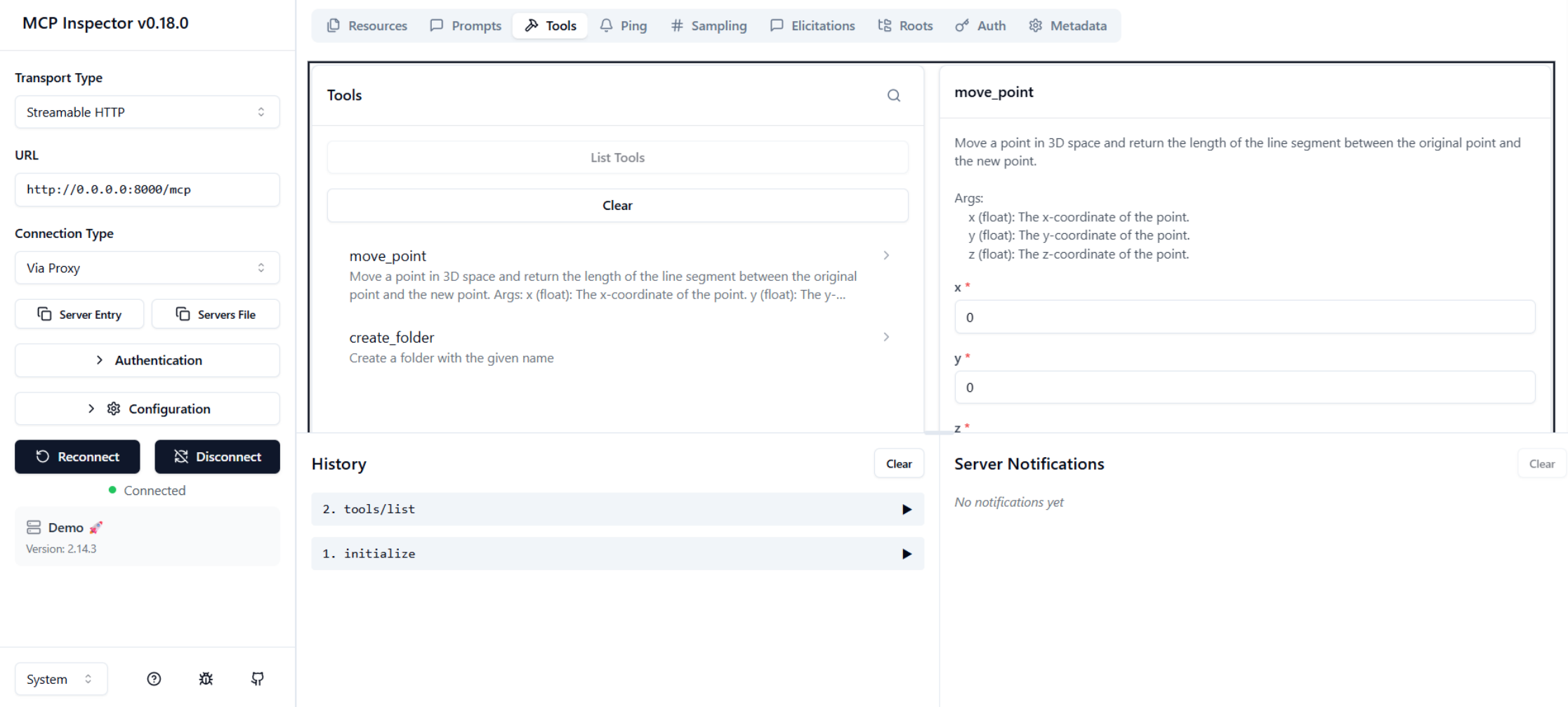
- Use of simulation plug-ins from grasshopper will be difficult.
- No learning about MCP (Model context protocol)
- Need Rhino 9(WIP) to use Python 3.10+
- Can so many so many nuances to learn.



03

Framework

**FastMCP > MCPserver
+ Any LLM UI**

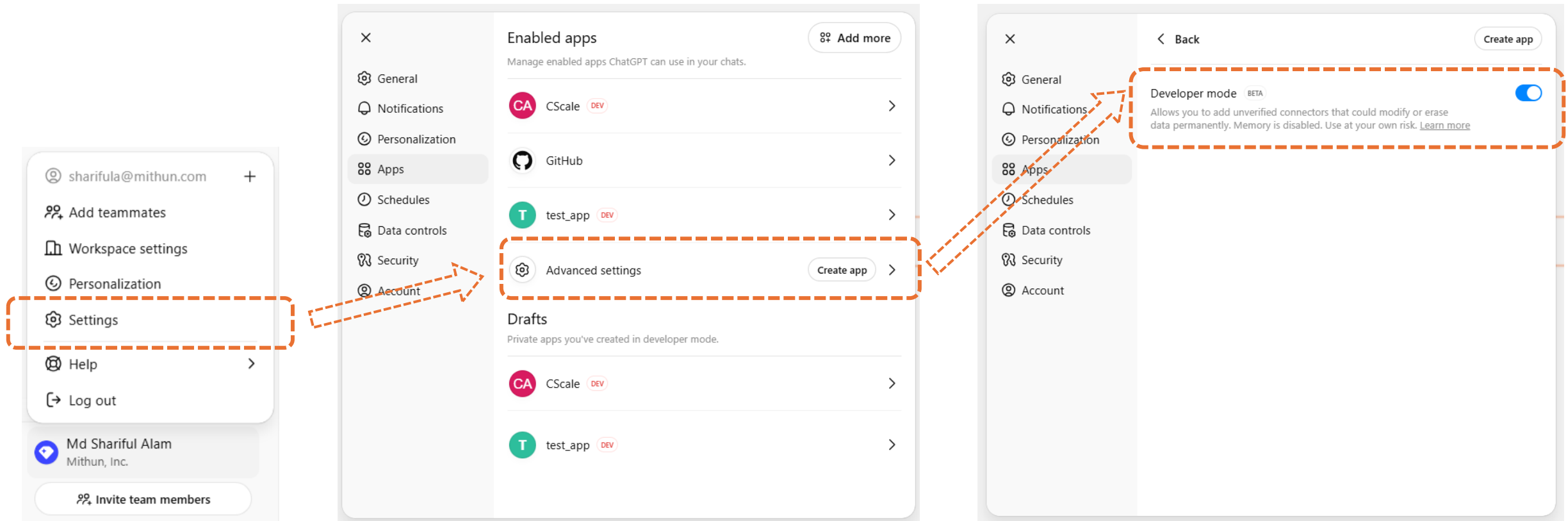


To open MCP inspector run this in your PowerShell:
`npx @modelcontextprotocol/inspector python mcp_server.py`

MCP Inspector

Steps to add MCP in ChatGPT

Framework *03*



Step 01: Turn on 'Developer mode' first

Steps to add MCP in ChatGPT



```
S:\Visualization\Projects\Cor X + - (Ctrl+C to quit)
ngrok

⚠ Free Users: Agents s3.19.x stop connecting 2/17/26. Update or upgrade: https://ngrok.com/pricing

Session Status      online
Account             AnikCOLAB (Plan: Free)
Update              update available (version 3.35.0, Ctrl-U to update)
Version             3.33.1
Region              United States (California) (us-cal-1)
Latency             36ms
Web Interface       http://127.0.0.1:4040
Forwarding           https://tetchily-insectean-lorina.ngrok-free.dev -> http://localhost:8001

Connections          ttl    opn    rt1    rt5    p50    p90
39                  0      0.00   0.00   5.02   5.25

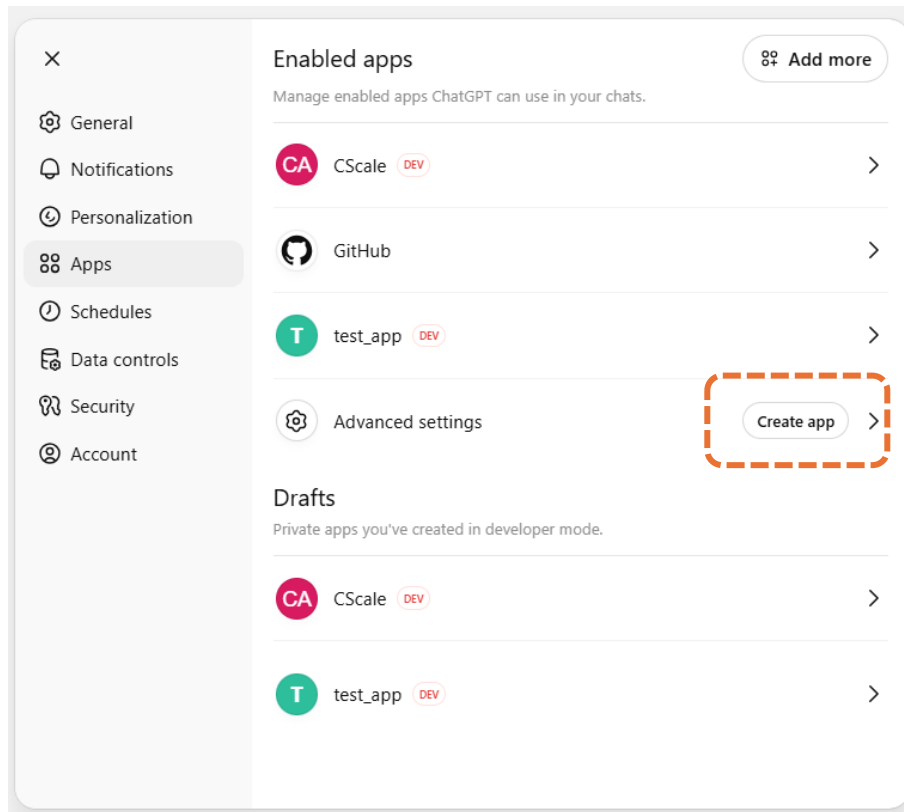
HTTP Requests
-----
17:55:52.122 PST POST /mcp          202 Accepted
17:55:52.246 PST POST /mcp          200 OK
17:55:52.090 PST GET /mcp           200 OK
17:55:51.967 PST POST /mcp          200 OK
17:53:06.219 PST GET /mcp           200 OK
17:53:06.059 PST POST /mcp          200 OK
17:53:06.328 PST POST /mcp          200 OK
17:53:06.214 PST POST /mcp          202 Accepted
17:52:17.269 PST POST /mcp          200 OK
17:52:17.708 PST POST /mcp          200 OK
```

Step 02: *Turn the local server to a public server*

In Ngrok type: **Ngrok http {you server can be 8000/8001}**

You will get this link: <https://tetchily-insectean-lorina.ngrok-free.dev>

Steps to add MCP in ChatGPT



Step 03: Create an App.
Paste the link with '/mcp': <https://tetchily-insectean-lorina.ngrok-free.dev/mcp>

Must add '/mcp' after
the link generated
from ngrok

New App BETA

+ Icon (optional)
Minimum size: 128 x 128 px

Name
test_app

Description (optional)
Explain what it does in a few words

MCP Server URL
https://tetchily-insectean-lorina.ngrok-free.dev/mcp

Authentication
No Auth

Custom MCP servers introduce risk. [Learn more](#)

☒ I understand and want to continue
OpenAI hasn't reviewed this MCP server. Attackers may attempt to steal your data or trick the model into taking unintended actions, including destroying data.

[Read the guide](#) **Create**

Developer mode

Memory is not used for this chat. [Settings](#)

+ Ask anything

📎 Add photos & files

🧠 Company knowledge

🖼️ Create image

🔍 Deep research

🌐 Web search

⋮ More

🧠 Company knowledge

🤖 Agent mode

🎨 Canvas

📖 Study and learn

CA CScale DEV

T test_app DEV

🧠 Your Year with ChatGPT

📄 Quizzes

🐙 GitHub

You can find the application
from '...more' in the '+' button

☰ Called tool ^

Framework 03

T test_app
move_point

Request

📄 Copy

> {x: 5, y: 5, z: 70}

Response

📄 Copy

> {result: 'Moved point from [5, 5, 5] to [5.0, 5.0, 70.0]. L...'}

✅ Done.

I moved the point **only in the Z direction** so that the line length is exactly **65 units**.

- Original point: (5, 5, 5)
- New point: (5, 5, 70)
- Resulting line length: 65 units

This keeps X and Y unchanged and adjusts Z by +65, which satisfies your requirement.

📄 👍 🗨️ ⬆️ ↺ ⋮

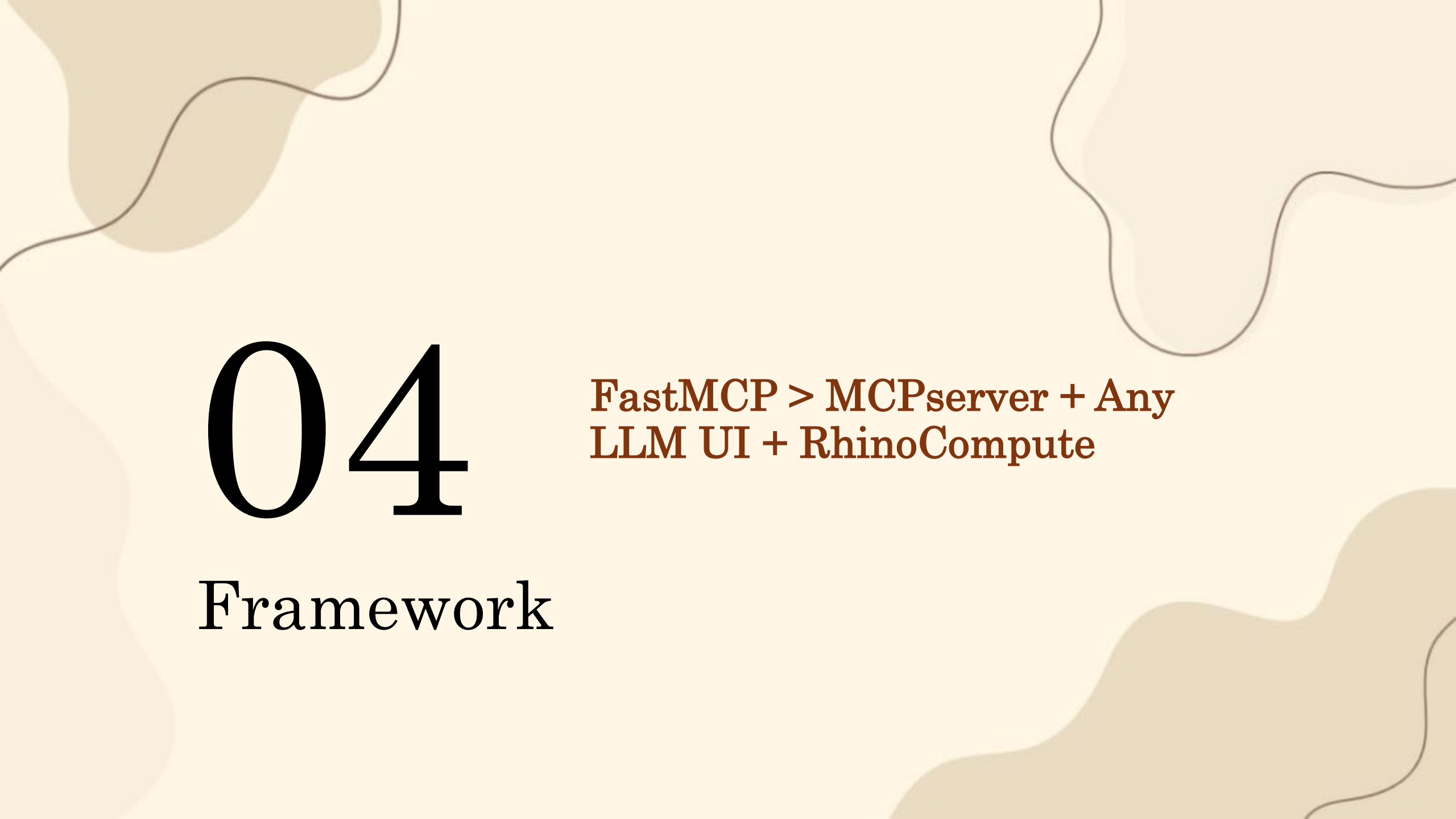
Framework *03*

Merits

- Get in depth knowledge about MCPs
- Fast and Easy to connect and test in any LLM platform.
- Easy to create and track small functions

Demerits

- No CAD environment connected
- Need Basic programming background and understanding of server connections.
- No Handoff concept.



04

Framework

**FastMCP > MCPserver + Any
LLM UI + RhinoCompute**

Framework 04

```
C:\Users\sharifula\AppData\Roaming\McNeel\Rhinoceros\packages\8.0\Hops\0.16.28\rhino.compute\rhino.compute.exe
RC [17:22:14 INF] Rhino compute started at 1/19/2026 5:22:14 PM
RC [17:22:14 INF] Now listening on: http://localhost:6500
RC [17:22:14 INF] Application started. Press Ctrl+C to shut down.
RC [17:22:14 INF] Hosting environment: Production
RC [17:22:14 INF] Content root path: C:\Users\sharifula\AppData\Roaming\McNeel\Rhinoceros\packages\8.0\Hops\0.16.28\rhino.compute
RC [17:22:18 INF] Starting compute.geometry instance on port 6001
CG [17:22:21 INF] Child process started at 1/19/2026 5:22:21 PM
CG [17:22:21 INF] Compute 8.0.0.0, Rhino 8.26.25349.19001
CG [17:22:33 INF] (1/4) Loading rhino commands plugin
CG [17:22:33 INF] Successfully loaded commands plugin
CG [17:22:33 INF] (2/4) Loading rhino scripting plugin
CG [17:22:33 INF] Successfully loaded scripting plugin
CG [17:22:34 INF] Configured scripting plugin for compute
CG [17:22:34 INF] (3/4) Loading grasshopper
CG [17:22:41 INF] (4/4) Loading compute plug-ins
CG [17:22:42 INF] Now listening on: http://localhost:6001
CG [17:22:42 INF] Application started. Press Ctrl+C to shut down.
CG [17:22:42 INF] Hosting environment: Production
CG [17:22:42 INF] Content root path: C:\Users\sharifula\AppData\Roaming\McNeel\Rhinoceros\packages\8.0\Hops\0.16.28\rhino.compute
RC [17:23:14 INF] Max concurrent requests = 0
RC [17:24:14 INF] Max concurrent requests = 0
RC [17:25:14 INF] Max concurrent requests = 0
RC [17:26:14 INF] Max concurrent requests = 0
RC [17:27:14 INF] Max concurrent requests = 0
RC [17:28:14 INF] Max concurrent requests = 0
RC [17:29:14 INF] Max concurrent requests = 0
RC [17:30:14 INF] Max concurrent requests = 0
RC [17:31:14 INF] Max concurrent requests = 0
```



To get RhinoCompute Running
install 'Hops' and open Grasshopper

RhinoCompute

Framework 04

MCP Inspector v0.18.0

Transport Type
Streamable HTTP

URL
http://0.0.0.0:8001/mcp

Connection Type
Via Proxy

Server Entry Servers File

Authentication

Configuration

Reconnect Disconnect

Connected

Simple MCP with Rhino.Compute
Version: 2.14.3

Resources Prompts Tools Ping Sampling Elicitations Roots Auth Metadata

Tools

List Tools

Clear

get_rhinocompute_version_details
Retrieves version information from the connected Rhino.Compute server. Use this tool when you need to verify that Rhino.Compute is running, check compatibility, or confirm the...

get_installed_rhino_plugins
Returns a list of Rhino plugins installed on the Rhino.Compute server. Use this tool when checking server capabilities, diagnosing missing plugins, or validating that a Grasshopper ...

get_installed_grasshopper_plugins
Returns a list of Grasshopper plugins installed on the Rhino.Compute server. Use this tool when checking server capabilities, diagnosing missing plugins, or validating that a...

read_grasshopper_inputs_outputs
Reads the inputs and outputs of a Grasshopper definition using Rhino.Compute's /io endpoint. Use this tool when you want to understand what parameters a .gh/.ghx file...

run_grasshopper_tool

History
Clear

3. tools/call

2. tools/list

1. initialize

read_grasshopper_inputs_outputs

Args:
pointer (str): Absolute or relative path to a .gh or .ghx Grasshopper file.

Returns:
dict: {
 "status": "success",
 "path": pointer,
 "description": "...",
 "inputs": [...],
 "outputs": [...],
}

pointer *
S:\Visualization_Projects\Computation\UW_class\Frameworks_example\Framework_04\hop_function_01.gh

Tool-specific Metadata:
No metadata pairs.

Add Pair

Output Schema:
Expand
{
 type: "object"

Server Notifications
Clear
No notifications yet

To open MCP inspector run this in your PowerShell:
`npx @modelcontextprotocol/inspector python mcp_rhinocompute.py`

MCP Inspector

Framework 04

Create an app in similar way
in ChatGPT

You should see all the
functions listed



General

Notifications

Personalization

Apps

Schedules

Data controls

Security

Account

Actions

get_installed_grasshopper_plugins

PUBLIC
WRITE

OPEN
WORLD

DESTRUCTIVE

Returns a list of Grasshopper plugins installed on the Rhino.Compute server. Use this tool when checking server capabilities, diagnosing missing plugins, or validating that a Grasshopper workflow will run correctly. Args: None Returns: dict...

METADATA

Visibility

public

get_installed_rhino_plugins

PUBLIC
WRITE

OPEN
WORLD

DESTRUCTIVE

Returns a list of Rhino plugins installed on the Rhino.Compute server. Use this tool when checking server capabilities, diagnosing missing plugins, or validating that a Grasshopper or Rhino workflow will run correctly. Args: None Returns: dict: {...

METADATA

Visibility

public

get_rhinocompute_version_details

PUBLIC
WRITE

OPEN
WORLD

DESTRUCTIVE

Retrieves version information from the connected Rhino.Compute server. Use this tool when you need to verify that Rhino.Compute is running, check compatibility, or confirm the Rhino/Compute build before running Grasshopper definitions. Arg...

METADATA

Visibility

public

read_grasshopper_inputs_outputs

PUBLIC
WRITE

OPEN
WORLD

DESTRUCTIVE

Reads the inputs and outputs of a Grasshopper definition using Rhino.Compute's /io endpoint. Use this tool when you want to understand what parameters a

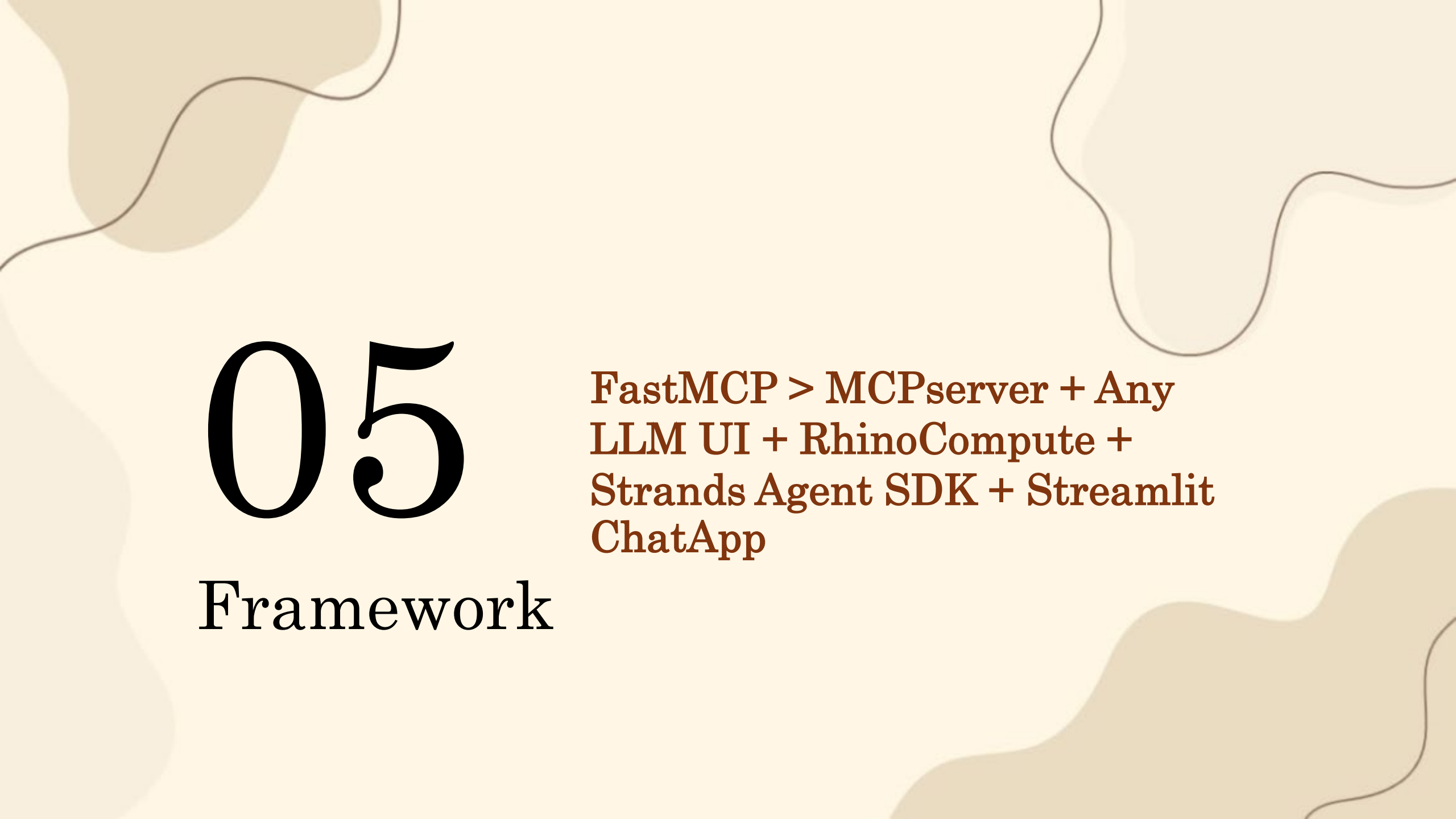
Framework *04*

Merits

- Large functions can be called with Rhino Compute, while using Grasshopper plug-ins
- Limitless possibilities in scaling up the system and framework

Demerits

- Have access to CAD environment but outcomes are not instantly visualized.
- Need Advanced programming background
- Installing Hops/ RhinoCompute can be an issue



05

Framework

FastMCP > MCPserver + Any
LLM UI + RhinoCompute +
Strands Agent SDK + Streamlit
ChatApp

Python API

[Agent](#)[Agent](#)[Agent Result](#)[Base](#)[Conversation Manager](#) >[State](#)[Event Loop](#)[Event Loop](#)[Streaming](#)[Handlers](#)[Callback Handler](#)[Hooks](#)[Events](#)[Registry](#)[Models](#)[Model](#)[Anthropic](#)[Bedrock](#)[Gemini](#)[Litellm](#)[Llamaapi](#)[Llamacpp](#)[Mistral](#)[Ollama](#)[Openai](#)

strands.agent.agent

Agent Interface.

This module implements the core Agent class that serves as the primary entry point for interacting with foundation models and tools in the SDK.

The Agent interface supports two complementary interaction patterns:

1. Natural language for conversation: `agent("Analyze this data")`
2. Method-style for direct tool access: `agent.tool.tool_name(param1="value")`

```
AgentInput = str | list[ContentBlock] |  
list[InterruptResponseContent] | Messages | None  
module-attribute
```

```
AgentState = JSONSerializableDict module-attribute
```

```
AttributeValue = Union[str, bool, float, int,  
List[str], List[bool], List[float], List[int],  
Sequence[str], Sequence[bool], Sequence[int],  
Sequence[float]] module-attribute
```



On this page

[AgentInput](#)[AgentState](#)[AttributeValue](#)[Messages](#)[T](#)[_DEFAULT_AGENT_ID](#)[_DEFAULT_AGENT_NAME](#)[_DEFAULT_CALLBACK_HANDL...](#)[logger](#)[AfterInvocationEvent](#)[should_reverse_callbacks](#)[Agent](#)[system_prompt](#)[tool](#)[tool_names](#)[__call__](#)[__del__](#)[__init__](#)[cleanup](#)[invoke_async](#)[stream_async](#)[structured_output](#)[structured_output_async](#)[AgentInitializedEvent](#)[AgentResult](#)[__str__](#)

Strands Agents and MCP servers

This demo shows how to use Strands to create an AI assistant and extend it with external tools.

Current Model: OpenAI - gpt-5



Explain me the input and output of this gh file:

S:\Visualization_Projects\Computation\UW_class\Frameworks_example\Framework_04\hop_f
unction_01.gh



Using tool: read_grasshopper_inputs_outputs

```
{  
  "pointer": "S:\\Visualization\\_Projects\\Computation\\UW_class\\Framework_04\\hop_f  
unction_01.gh"  
}
```

Tool Result: success

```
{"status": "success", "path": "S:\\Visualization\\_Projects\\Computation\\UW_c
```

Here's what that Grasshopper file expects and returns:

Inputs

Ask your agent...



Libraries used:

- Strands Agent
- Streamlit
- FastMCP
- RhinoCompute

Server list:

1. RhinoCompute Server
2. MCP server
3. Streamlit Application server.

Framework *05*

Merits

- Any LLM can be tested and used in the framework
- Deployment level learning, the current state-of-the-art Agentic workflow eco-system
- 3D changes can be visualized within the platform

Demerits

- Need to learn multiple python libraries: Strands Agent, Streamlit
- Should have strong background in calling APIs

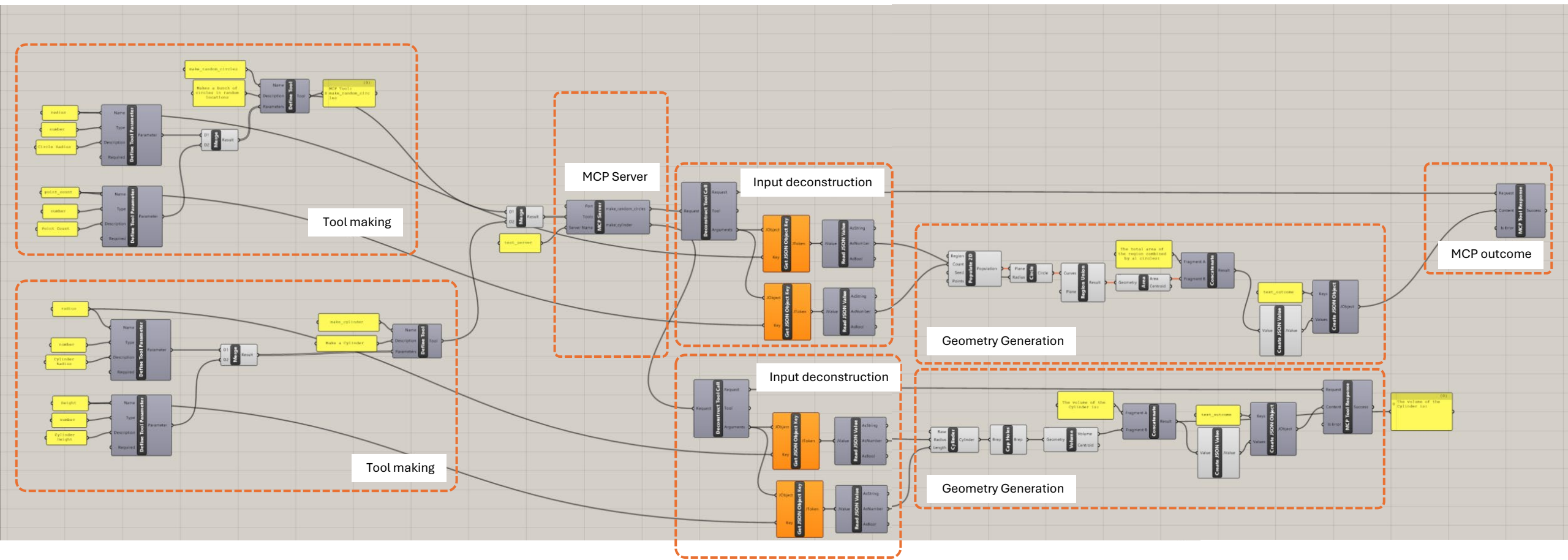


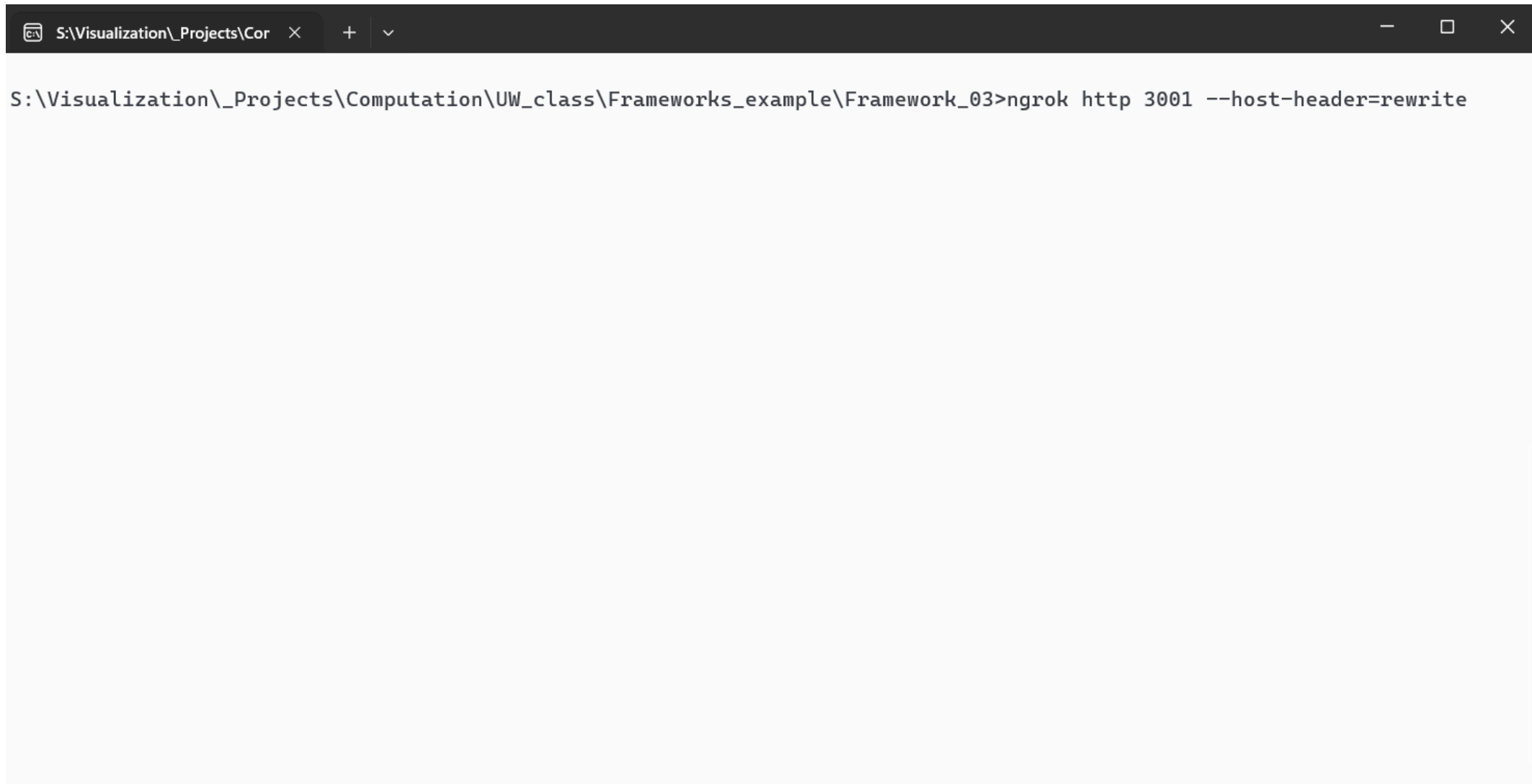
06

Framework

Grasshopper + Swiftlet
+ Any LLM UI

Framework 06



A screenshot of a terminal window with a dark title bar. The title bar contains a file icon, the path 'S:\Visualization_Projects\Cor', and window control buttons (close, maximize, and a dropdown arrow). The terminal area is light gray and contains the command 'S:\Visualization_Projects\Computation\UW_class\Frameworks_example\Framework_03>ngrok http 3001 --host-header=rewrite' in a dark monospaced font.

```
S:\Visualization\_Projects\Computation\UW_class\Frameworks_example\Framework_03>ngrok http 3001 --host-header=rewrite
```

In Ngrok you must add additional (`--host-header=rewrite`):
`ngrok http 3001 --host-header=rewrite`

Framework ***06***

MCP Inspector v0.19.0

Transport Type
Streamable HTTP

URL
https://tetchily-insectean-lorina.ngrok-free.dev//mcp

Connection Type
Via Proxy

Server Entry Servers File

Authentication

Configuration

Reconnect Disconnect

Connected

test_server
Version: 1.0.0

Resources Prompts Tools Tasks Ping Sampling Elicitations Roots Auth Metadata

Tools

List Tools

Clear

make_random_circles
Makes a bunch of circles in random locations

make_cylinder
Make a Cylinder

make_random_circles

Makes a bunch of circles in random locations

radius *

6

point_count *

8

Tool-specific Metadata: Add Pair

No metadata pairs.

Run as task

Run Tool

Copy Input

Tool Result: Success

```
{
  text_outcome: "The total area of the region combined by al circles: 529.10
5887"
}
```

History Clear

3. tools/call

2. tools/list

1. initialize

Server Notifications Clear

36. notifications/message

35. notifications/message

34. notifications/message

33. notifications/message

32. notifications/message

System

To open MCP inspector run this in your terminal then:
npx @modelcontextprotocol/inspector

Framework ***06***

What are the tools available in GH app?

Looked for available tools

Here's what's currently available in the **GH App** 📌

(I pulled this directly from the app, so this is the full, up-to-date list.)

Available GH App Tools

1 Make a Cylinder

Tool path:

`/GH App/.../make_cylinder`

What it does:

Creates a cylinder and computes its volume.

Inputs:

- `radius` → Cylinder radius
- `Height` → Cylinder height

Output:

- Volume of the cylinder

2 Make Random Circles

Tool path:

`/GH App/.../make_random_circles`



W

Ask anything



 GH App



make an app in
ChatGPT like before

Framework *06*

Framework *06*

Merits

- Have access to CAD environment and outcomes are instantly visualized.
- Will give students basic ideas about MCP with no Programming knowledge

Demerits

- Agent Handoffs/swarm not implemented



Thank you